

CCM Plant and Auxiliaries Operating Manual

For the Centre for Comparative Medicine

4145 Wesbrook Mall



Building Operations

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Clayton Boiler – New OIU Interface

Procedures – Removed from this manual

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Introduction and Operator Requirements

Introduction

It is the intention of the operations manual to provide site specific information to a competent power engineer in understanding, maintaining, and skillfully operating the CCM steam plant.



Figure 1 - Artist's rendering of the CCM facility.

The Centre for Comparative Medicine (CCM), located at the southern end of the University of British Columbia Point Grey campus, is a 60-thousand-square-foot facility for comparative medicine research. The CCM is owned and operated by UBC Building Operations.

The complex consists of single story buildings constructed mainly from concrete, masonry block and steel.

The CCM plant steam boilers produce saturated 100 PSig steam for both heating and process use. Steam from the boiler header is stepped-down in pressure through pressure reducing stations and

fed into three distribution headers.

These headers provide steam to the facility's operating equipment (sterilizers and digesters), domestic-hot-water heating systems, and the facility's HVAC systems.

The load-demand for process steam varies (swings) depending on the facility's day-to-day operations.

The CCM plant also houses a bank of heat pumps and two hot water boilers provide water for heating, cooling and domestic water use.

Location

The formal address of the CCM facility is 4145 Wesbrook Mall, Vancouver BC. The location can be seen on the map on the next page, marked by the black arrow.

The CCM plant is housed in the front section of the building, to the right of the offices, reception, and lobby.

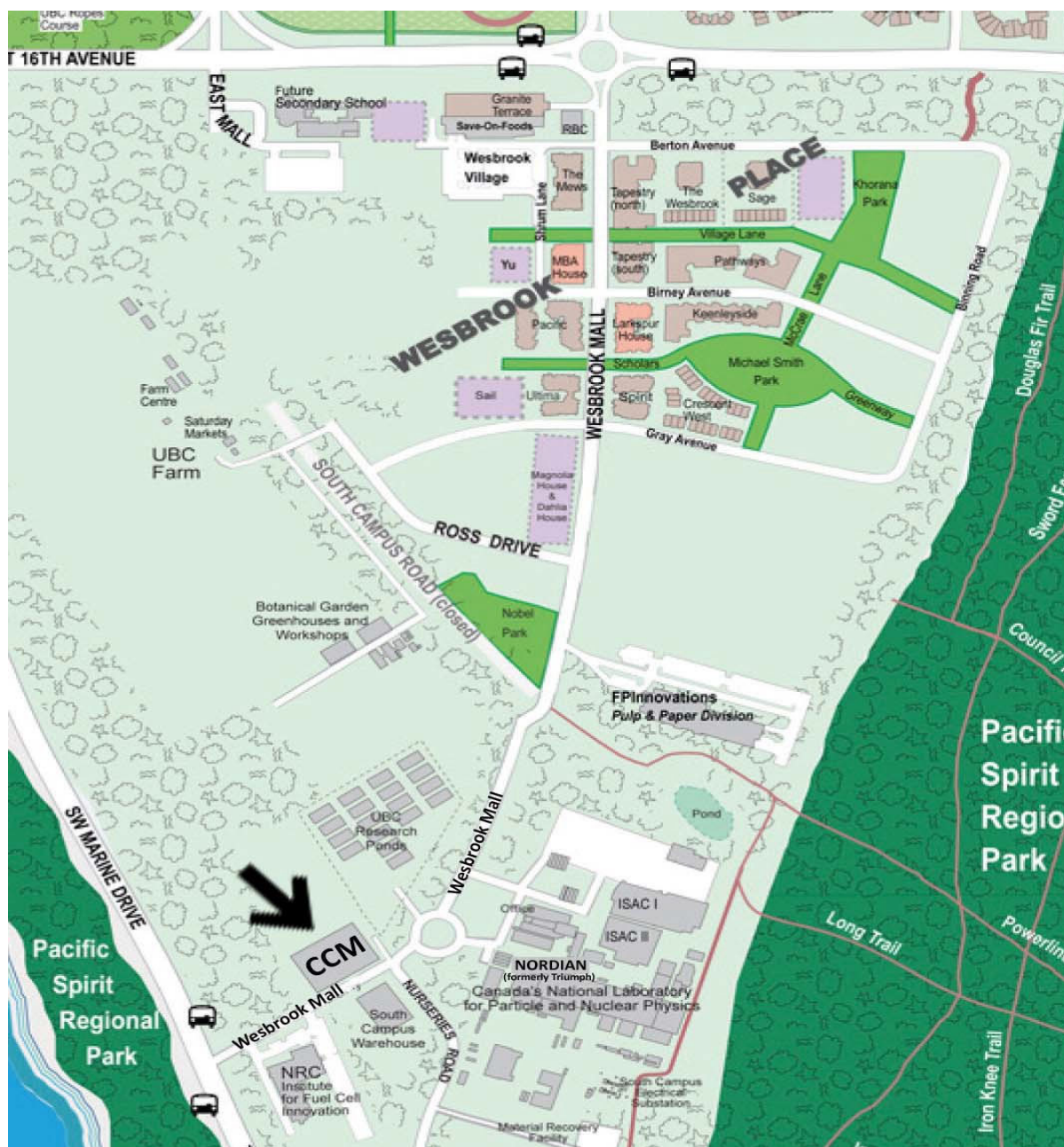


Figure 2 - The Location of the Centre for Comparative Medicine

Operator Requirements and Assumptions

This manual is written with the assumption that the operators of the CCM plant are skilled power engineers with at least a fourth-class certification from the BC Safety Authority.

While this manual covers the basic operations of the plant, it is understood that the power engineer in charge of the plant has the autonomy to make decisions in the best interest of the safe operation of the plant.

Before taking charge of the plant, the power engineer shall demonstrate their competency by passing a written examination covering the fundamentals of operating the CCM steam plant and by demonstrating their knowledge by skillfully operating the plant equipment in the presence of a qualified CCM steam plant operator.

Related Phone Numbers

CCM Facility Manager – 604-822-6818

UBC Chief Engineer – 604-827-0038

UBC CEC Control RM – 604-822 8512

UBC BRDF Control RM – 604-827-1006

CCM Steam Plant Boiler Room – 604-827-0071

CCM Reception – 604-827-4938

Rules, Regulations, and Other Reference Guides

In addition to this manual, the power engineer is required to be familiar and comply with the following:

Regulations and Variances

- The BC Safety Standards Act
- The Power Engineers, Boiler, Pressure Vessel and Refrigeration Safety Regulation. (the BC Boiler regulation)
- The *Workers Compensation Act*, specifically the Occupational Health and Safety Regulation
- CSA B51 and B52
- All variances (i.e. Alternative Safety Approach) in effect at the CCM

UBC Policies, Standing Orders and Procedures

- UBC Safety Procedures and Lockout Procedures
- UBC Specific CCM Procedures Booklet

Manufacturers' Manuals

The manufacturer's manuals for the main equipment in the plant as listed below:

- The Clayton Industries Steam Generator & Fluid Heater Installation Manual
- The Clayton Industries Steam Generator Instruction Manual
- The Clayton Industries Feedwater Treatment Manual
- The 3200NXT Water Softener Controller Manual
- The Cleaver-Brooks Model FLX Packaged Boiler Operation, Service and Parts Manual
- The Cleaver-Brooks Hawk 1000 Boiler Controls Manual
- The Cleaver-Brooks Level Master CBLME Water Level Control Manual
- The Grundfos CR, CRI, CRN, CRT Installation and Operating Manual

Mechanical Drawings

- M6.05A – Steam and condensate system P&ID
- MCW drawing M6.05 – Boiler room P&ID
- UBC Centre for Comparative Medicine - Record Drawings (2016)

Stable Plant Operation

Tools and Personal Protective Equipment

Before entering the plant, it is required that the following personal protective equipment (PPE) is available and on hand:

- Hearing protection (ear plugs and/or acceptable ear muffs)
- Leather and chemical proof gloves
- Eye protection
- Long-sleeved shirt
- CSA approved safety boots (steel-toed or acceptable composite)

For feedwater chemical handling and mixing, the following is suggested to be worn in addition to the above PPE:

- A chemical-resistant apron
- Chemical-resistant gloves
- A face shield

Following Procedures

The Standard Operating Procedures (SOPs) in this manual are to be followed in the exact sequence outlined. Any deviation or change to the procedures must be pre-approved by the UBC Campus Chief Engineer.

Equipment and Process Overview

Steam Boilers

As of August, 2016, the CCM steam plant houses a Cleaver-Brooks Model FLX (Flextube) packaged water-tube boiler (C.B. Boiler) and a Clayton Industries one-through coil-tube steam generator (which is referred to as the Clayton Boiler herein in this document).

The Clayton Boiler is designated as Boiler Number One and the C.B. Boiler is designated as Boiler Number Two. It is normal practice for the C.B. Boiler to act as the lead boiler with the Clayton Boiler as the stand-by (lag) boiler.

These boilers widely differ in operation and have their own sets of operating procedures. Each boiler has its own section in this guide.

The main features of the Clayton Boiler are:

- Is a once-through coil-tube boiler with no steam space;
- Water does not circulate through the boiler. It is forced through;
- Steam pressure is determined by the feedwater pump discharge pressure;
- Can quickly generate steam from cold start, though it is imperative to not overheat the coil;
- Can quickly meet changes in load;
- Water in the produced steam is knocked out by cyclone-action in the mechanical steam separator. The water is returned to the condensate tank via a trap;
- Safeguards on this boiler include high-separator-level and high-coil-temperature, in addition to common boiler safe guards;
- The feedwater pump is a positive displacement pump.

The main features of the C.B. Boiler are:

- Is a flex-tube water-tube boiler; model FLX-700-300-150 ST;
- Steam pressure is determined by the boiler firing rate, creating pressure in the steam drum;
- Has a mud- and steam-drum, allowing the boiler to have a steam space;
- Can handle low-load demands better than the Clayton Boiler;
- The gauge-glass, on the steam drum, must normally show a level of about 50 percent. A full and empty gauge glass must be avoided;
- Is slow to generate steam from cold start (requires between two to four hours of warming up);
- Water in the boiler circulates through the boiler by natural convection;
- Trips on low- and high-drum-level, in addition to common boiler trip conditions;
- The feedwater pump is a centrifugal pump, providing water to the boiler above set point pressure

Steam Distribution System

The CCM facility contains one boiler steam header and three steam distribution headers. The power engineer must understand the following steam distribution systems and the associated pressure reducing stations prior to taking responsibility for the plant:

- **135 PSI system** - steam at nominally 100 PSIG is fed only to the main steam header for let down into the 12 PSIG, 45 PSIG and 80 PSIG steam systems. The 100 PSIG system is protected by the safety valves on the boilers.
- **12 PSI system** - supplies the humidification system, two domestic water heaters, hydronic system heaters, and the air handling units. Maximum flow on the 12 PSIG system is 12,584 lbs. /hour. The 12 PSIG system safety valve is set at 15 PSIG.
- **45 PSI system** - supplies the Sterilizer in Room 1060 and the tunnel cage washer in room 1133. The Sterilizer has a maximum steam demand of 1,080 lbs. /hour. The tunnel washer has a maximum steam demand of 1,500 lbs. /hour. The 45 PSIG system safety valve is set at 50 PSIG.
- **80 PSI system** - supplies the cage rack washer in room 1133 and the tissue digester in room 1117A. The cage washer has a maximum steam demand of 60 lbs. /hour. The tissue digester has a maximum steam demand 325 lbs. /hour and 280 lbs. / cycle. The 80 PSIG system safety valve is set at 90 PSIG.

Steam Plant Control Philosophy

Pressure in the plant's main boiler steam header is controlled by the pressure transmitter and firing rate of the designated lead boiler.

If the two steam boilers are online at the same time, each boiler will operate off their own pressure transmitters to maintain their own set points. This may cause plant operating problems as the Clayton Boiler has a higher steam capacity than the C.B. Boiler.

Therefore, only one steam boiler in the CCM steam plant should be in operation, with the other in auxiliary standby mode. An exception to this is when the operator is making a transition from one boiler to another.

When the Clayton Boiler is designated as the auxiliary standby boiler, it will monitor the plant using a pressure transmitter on the main-steam header. If the main-steam header pressure drops to a preset value, the Clayton Boiler will automatically come online to prevent a major drop in plant steam pressure.

The readings from the pressure transmitter on the main-steam header are also logged on the DDC and can be trended by the operator.

In addition to this pressure transmitter, there is a steam flow transmitter which has its readings logged on the DDC system. These readings can also be trended by the operator.

Plant Natural Gas Supply

The CCM facility is fueled by natural gas supplied by FortisBC. The gas metering station is located outside in the south-east corner of the CCM compound.

The gas meter station has a seismic shutoff valve, a series of pressure-reducing valves, and a gas integrator. It is the responsibility of the operator to record the reading on the integrator at the end of their shift to document the facility's fuel consumption.

Event Recording and Operator Checks

Log Book Entries

The CCM Log Book is located in the steam boiler room, on the desk beside the DDC monitoring station.

The Log Book is to be filled out as per CCM Plant Operating Procedure #16.

Keep entries in the Log Book clear and concise. Avoid the use of unfamiliar acronyms and keep terminology consistent among operators. A copy of a page from the Log Book is featured below:

UBC CCM Steam Plant											
Daily Operating and Maintenance checks											
Date:											
		Steam Pressure		Firing rate (0 - 100%)		Stack Temp.		Gas supply pressure (psi)		Flame signal (0 - 100%)	
#1 Boiler											
#2 Boiler											
Make-up water integrator				Blowdown hotwell gage glass							
Softener integrator				Blowdown booster pump strainers							
Hotwell level				Blowdown hotwell float chamber							
Hotwell temperature				#1 Feedpump pressure							
Booster pump pressure				#2 Feedpump pressure							
Gas meter integrator (LCD)				Separator steam trap operation OK?							
Boiler Water Test Results										Condensate	
Time	TDS PPM	SO3 PPM	pH	P. Alk PPM	Chlorides PPM	M. Alk PPM	OH Alk PPM	Phosphate PPM	Hardness PPM	pH	TDS
Chemical additions											
Time	Sulfite (ounces)	Caustic (ounces)	Phosphonate (ounces)	Amine (ounces)	Chemical pump settings	Salt (lbs)					
					#1 =						
					#2 =						
Engineer's notes, observations, actions taken and equipment changes											
		First name					Last name				
Engineer											
Engineer											

Figure 3 - CCM Plant Log Book Page

Daily Shift Checks

The following table outlines the checks to be done by an operator in charge of the CCM plant on every shift. These recommended checks rely on the operator's experience and intuition. Remember to use the four senses (sight, touch, smell and sound) when assessing the plant.

Note: All auxiliary equipment (hot water boilers, heat pumps, domestic water systems, fire systems and the emergency generator) are considered to be part of the plant and are included in the responsibilities of the CCM plant operator.

For all scheduled maintenance procedures on equipment, consult the accompanying manufacturers' manual for additional information on all steps, tools, and safety procedures.

All daily checks are to be recorded as per The CCM Plant Log Book Entries Guide 2016.

Location	Operating Checks to Be Carried Out
CCM Steam Plant Boiler Room	a. Review past entries in the log book, paying attention to equipment outages and/or lockouts, unusual operations, and the results of past water tests. b. Perform a preliminary walk through of the steam boiler room, specifically checking the water levels and pressures of the operating steam boilers and that ample feedwater is available, which is at least a 50 percent level in the hotwell indicated by the hotwell gauge glass. c. During the walk around, look all around for leaks or malfunctioning equipment throughout the plant. Be sure to check motor-driven equipment for vibration and high temperatures. d. If you find a condition that puts the plant and lives in immediate danger, take action to remedy it immediately. e. Check the DDC monitoring system, looking for any unusual process values. f. Take readings as required for the log book. g. Perform feedwater tests and feedwater chemical batching as prescribed by the CCM plant's water treatment plan. h. Perform blow downs on boiler gauge glasses. For the C.B. Boiler, blow down the Level Master Water Level Controller as per the Level Master manual. i. Manually blow down the Clayton Boiler and/or the Cleaver-Brooks as per the feedwater treatment manual.

Location	Operating Checks to Be Carried Out
CCM Heat Pump, Air Compressor and Suction Systems (Upper Mezzanine)	a. Perform a walkthrough of the heat pump, compressed air system, vacuum (suction) air systems and auxiliary systems. b. Check the cooling tower chemical pump and chemical supplies. c. Check all pumps for bearing vibration, high bearing temperatures.
CCM Cooling Tower Pumps, Motor Control Cabinet, Fire Systems, and Domestic Water	a. Perform a walkthrough of the heat pump, air system, and auxiliary systems. Look for any equipment displaying error messages and take appropriate action. b. Check the Motor Control Cabinet for abnormal breaker positions. Call an electrician if there any problems. c. Check the cooling tower water pumps and auxiliaries (i.e. air separator) for proper operation. Check pumps for bearing vibration, high bearing temperatures, and leaks. d. Visually check the fire system pressures. If there are any major deviances, contact an available supervisor immediately. e. Visually check the raw water line entering the CCM is not leaking and that the backflow preventer is not leaking.
Hot-Water Boiler Room	a. Walk through the hot-water boiler room and look for system leaks and look for any equipment displaying error messages and take appropriate action. b. Verify that the hot water boilers are operating as intended. Troubleshoot as per the manufacturer's recommendations, detailed in the manufacturer's manual.
Emergency Generator Room	a. Inspect the generator room, looking for oil, fuel, and coolant leaks from the generating unit. b. Visually check the starter-motor battery banks for corrosion, leaks, and wear. c. Report any condition that may impede the proper performance of the emergency generator.

Weekly, Monthly, Yearly Checks

In addition to the shift checks, there are weekly, monthly and yearly checks that have to be carried out. The following tables summarize these checks which must be performed to maintain operation and safety requirements.

Interval	Operating Checks to Be Carried Out
Weekly Checks	a. Blow down the Hotwell (feedwater) tank for 20 seconds. Also blow down the Hotwell gauge glass and water column. b. Blow out the strainers for the condensate return tanks. Ensure proper lockouts are in place when opening process piping as per the UBC Lockout Procedure. c. Blow out the strainers on the two booster pump suction lines. d. Check motor-driven equipment for lubrication and abnormal vibration. e. Check condensate, booster and feedwater pump as per manufacturer's instructions. f. Verify the plant valves are positioned properly (refer to the P&IDs for assistance). g. Verify the boiler controls and set points are properly set. h. For the Clayton Boiler: • Test the high-water-level switch on the separator. This is done by having the boiler online and closing the condensate drain valves on the separator (including the valve downstream of the separator steam trap). This will trip the boiler. Re-start the boiler as outlined in the start-up procedure in this manual. i. For the C.B. Boiler • Blow down all water columns and test LWCO switches. This will trip the boiler. Re-start the boiler as outlined in the start-up procedure in this manual.

Interval	Operating Checks to Be Carried Out
Monthly Checks	a. For the Clayton Boiler: • Check and document in the log book the Clayton Boiler coil pressure and stack temperature at high-fire. • Flush the Clayton Boiler's feedwater pump discharge head. • Check "Dual element thermocouple sensor" and "Main temperature limit controllers" as detailed in Clayton Steam Generator Instruction Manual on pages 6-7 & 6-8. • Check Auxiliary Temperature Switch (ATS) setting as detailed in Clayton Steam Generator Instruction Manual on page 7-23. • Check belt tension on feedwater pumps and Forced-draft fans. • Blow down the pressure sensing line. This is the line that is common to the pressure trip switches on the front of the Clayton Boiler. It is recommended to do this when boiler is on standby to prevent an interruption in steam service and to not shock the instrumentation on this line. • Perform a coil wash as per the Clayton Boiler Manual. b. For both boilers and the rest of the plant: • Test all permissive and interlocks on the boilers (list found in appendices). • Swap over the condensate booster pumps. • Lubricate equipment and valves as per manufacturer's instructions and best practice. • Test the boiler emergency shutdown buttons. • Test operation of the Emergency Diesel Generator. See "The CCM Plant Generator Testing Procedure".

Interval	Operating Checks to Be Carried Out
Yearly Checks	a. Plant General: • Service and repair steam traps, strainers, and valves (as required). • Service the strainers on the booster pumps and on the condensate return tank. • Coordinate annual service by Pacific Boiler. • Test and service all safety valves and ensure adequate records are kept. • Check and clean all boiler fan impellers for the Clayton and C.B. Boilers. Refer to boiler maintenance manuals. • Rod out the Hotwell tank level control pipe and valve connections. • Assist with the annual inspection by the BSCA Boiler Safety Officer. Ensure lockouts are in place. • Inspect all air, fuel and other control linkages for wear, screw tightness and free movement. • Check burner photocells as outlined in each boiler's maintenance manual. Test the signal strength and replace where necessary. b. For the Clayton Boiler Feedwater Pump: • Inspect the check valves. • Change oil in the pump crankcase

CCM Feedwater System Checks

Feedwater system overview

The boiler feedwater comes from the plant Hotwell tank. The Hotwell tank features a steam sparging pipe to pre-heat the feedwater.

The Hotwell tank receives condensate returning from the process and HVAC systems throughout the facility. The condensate collects in a condensate collection tank located under the plant Hotwell tank. It is then transferred to the Hotwell via two condensate pumps that operate on a float-level switch.

One condensate pump is in service while the other is in standby. The standby pump is to be routinely tested and the pumps switched over as per procedure.

Condensate is handled by a variety of steam traps located throughout the facility and the steam plant. However, not all of the condensate returns and the system requires make up water.

Makeup water is provided by the facility's domestic water system and is conditioned by a pair of water softeners, which remove almost all of the hardness from the water, before being fed into the Hotwell tank. The amount of makeup water entering the feedwater system is controlled by a float-operated valve on the side of the Hotwell tank. There is also a gauge glass on the Hotwell tank that allows operators to verify the Hotwell tank level.

From the Hotwell tank, the feedwater is fed to two booster pumps which provide a minimum suction head pressure (around 12 PSIG) to the boiler feedwater pumps.

The suction lines for the feedwater pumps are interconnected but their discharge lines are not. Feedwater pump one directly feeds the Clayton Boiler and feedwater pump two feeds the C.B. Boiler. Each pump has its own unique operating characteristics:

The Clayton Boiler Feedwater Pump

The Clayton Boiler feedwater pump is Clayton-Industries proprietary positive-displacement pump. Its speed is controlled by a variable frequency drive controlled by the Clayton Boiler.

The speed of the pump is proportional to the firing rate of the Clayton Boiler. When the firing rate of the Clayton Boiler increases, the pump's speed will increase as well.

Note: If the Clayton Boiler is ever starved of water during firing, the coil tube will quickly overheat and become damaged. The high-coil-temperature switches should trip the boiler before major damage occurs. See Appendix B for more information.

The Clayton feedwater pump also operates intermittently when the Clayton Boiler is on standby. It does this to circulate boiler chemicals through the coil tube to prevent corrosion and sediment buildup. As noted elsewhere in this manual, the boiler should be placed into "dry standby" if running the feedwater pump is not an option. See the "The Clayton Industries Steam Generator Instruction Manual" for more information about placing the boiler into "dry standby".

The C.B. Boiler Feedwater Pump

The C.B. Boiler feedwater pump is a Grundfos CR3-12K multi-stage centrifugal pump. Unlike the Clayton Boiler's feedwater pump, the CR3-12K pump intermittently operates in an on-and-off fashion.

It is controlled by the Cleaver-Brooks' Level Master controller and only comes on when the drum level in the C.B. Boiler drops to a pre-determined set point programmed in the Level Master Controller. The pump then shuts off when the drum level in the Cleaver-Brooks increases to a pre-determined set point programmed in the Level Master controller.

For more information on the operation and proper care of the CR3-12K pump, refer to The "Grundfos CR, CRI, CRN, CRT Installation and Operating Manual"

CCM Water Treatment Parameters

Feedwater treatment for the CCM steam plant is critical and warrants its stand-alone section and operating procedures.

DANGER: The Clayton Boiler **MUST NOT OPERATE** using feedwater with hardness! The Clayton boiler coil tube cannot tolerate any hardness. Any hardness will quickly scale, plug and damage the coil tube.

Raw Water Treatment

As mentioned earlier, raw water is treated by a pair of softeners before entering the feedwater storage tank.

The plant operators must ensure the brine tank is maintained with water and salt and be well-versed in the procedure to refill the brine tank. Additional details on the softeners can be found in "The 3200NXT Water Softener Controller Manual."

Boiler Feedwater and Condensate Parameters:

Refer to the feedwater treatment manual for the current boiler feedwater parameters, list of chemicals, and proper dosing procedures.

Clayton Coil-Tube Boiler Overview

Feedwater and Steam Flow

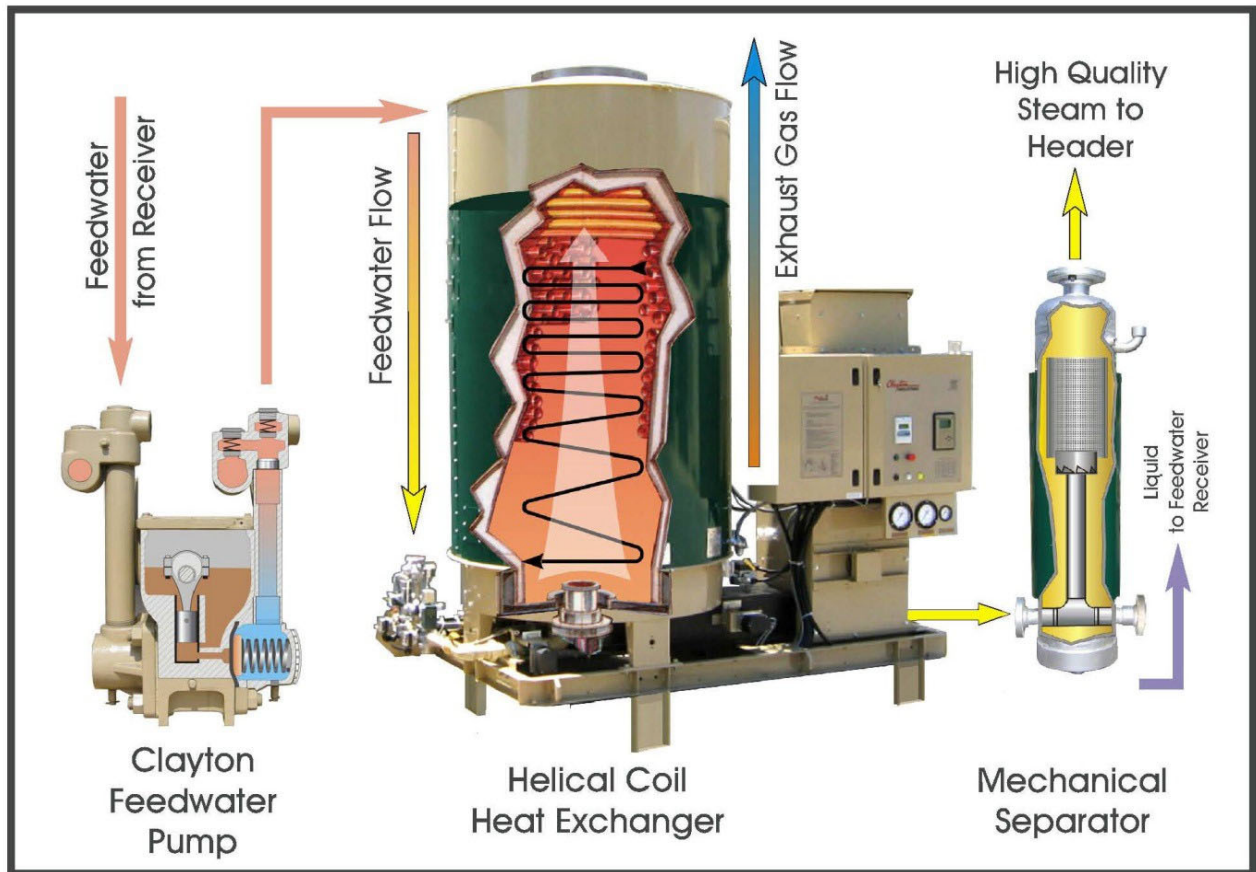


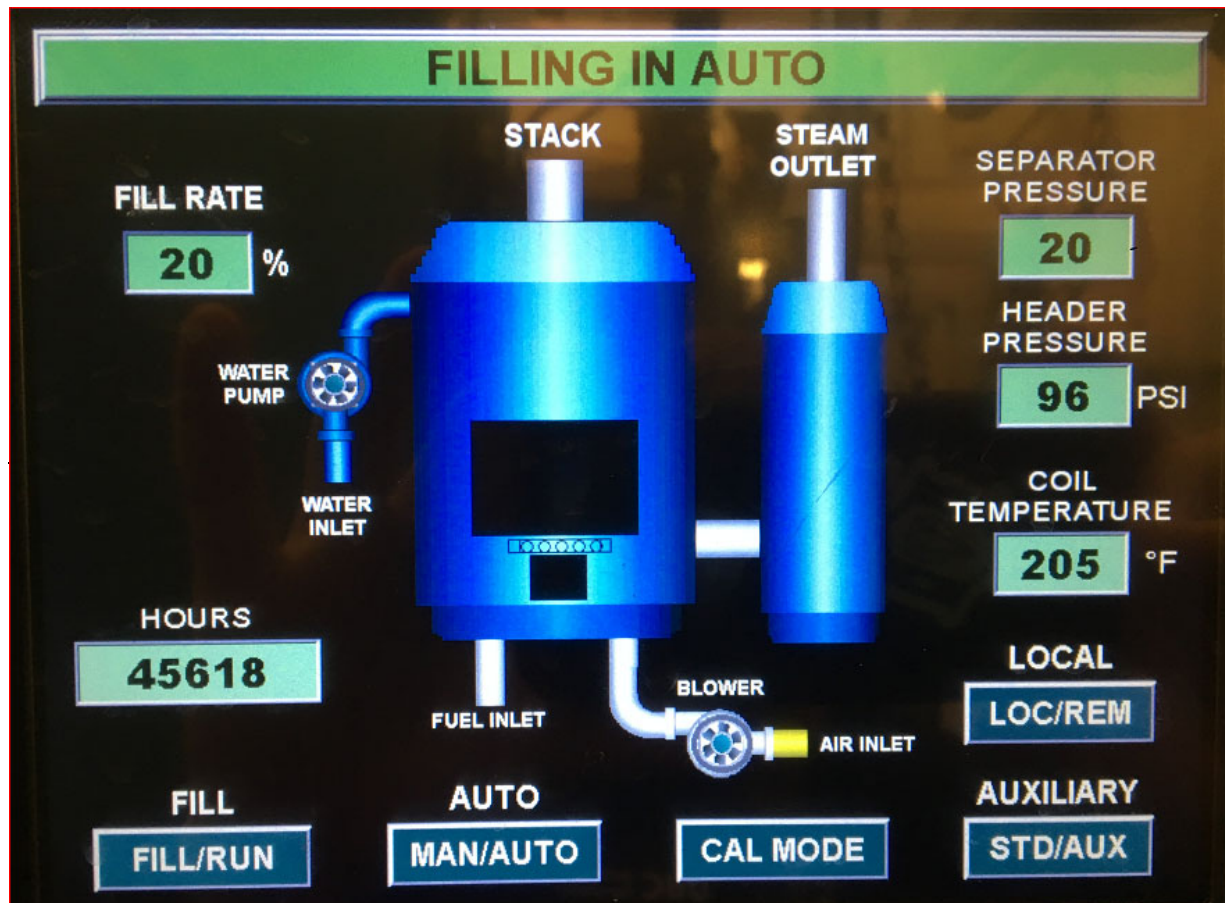
Figure 5 - Water and steam flow through a Clayton Boiler

As mentioned in the Steam Plant Operation, Equipment and Process Overview, section of this guide, the Clayton boiler is a once-through coil-tube boiler with no steam space. The steam pressure is determined by the speed of the positive-displacement feedwater pump. The speed of the pump is proportional to the boiler firing rate. The water travels through the Clayton Boiler in a single path in which it evaporates into steam. There is no natural recirculation in this once-through boiler, as there is in the Cleaver Brooks boiler.

The wet steam then enters the mechanical separator which knocks out any condensate in the steam. The result is a steam purity of about 99 percent leaving the separator and entering the plant's boiler steam header.

Clayton Boiler – Operator User Interface (OUI)

The Clayton Boiler has been fitted with a new operator user interface which is touch screen and can be operated by toggling the buttons on the screen.



Buttons on Touch Screen

FILL RATE: This shows the percentage that the positive displacement pump is running at. When the boiler burner is on, this is also the firing rate of the boiler.

HOURS: This shows the total accumulated amount of boiler run hours.

FILL/RUN: This toggle button is used to run the feed water pump in fill mode, and when toggled to run it starts the firing sequence.

MAN/AUTO: This toggles between manual and automatic firing control.

CAL MODE: This is a service button which is only for the service technicians.

STD/AUX: This is the button that controls whether the boiler operates under the control of the separator pressure sensor, or the header pressure sensor. The boiler should be left on AUX as it is able to control more accurately to changes in the header pressure

COIL TEMPERATURE: This is the coil temperature.

HEADER PRESSURE: This is the pressure sensor on the main header. This sensor is located on the header above the Clayton boiler

SEPARATOR PRESSURE: This is the pressure sensor on the separator. This sensor is located on the Clayton Boiler steam separator.

Starting the Boiler

1. To start the boiler turn on the main power switch on the A-Pillar and pull out the red Stop Button.
2. Once the PLC OUI boots up you will see the screen above.
3. Press the FILL/RUN button to start the firing sequence.
4. The boiler will fire up and run at 25% in MANUAL mode

Re-Starting the Clayton Boiler after a Power Outage

1. If the steam system is still pressurized, start at step 14 in the Clayton Boiler Start Up Procedure.
2. If the power outage is a longer duration and the steam system has fully de-pressurized, it must be slowly re-pressurized and a full start-up procedure must be executed

Clayton Industries Documentation

This section provides references to the Clayton Industries documentation pertaining to the CCM steam systems.

Clayton Industries Feedwater Treatment Manual

This manual contains essential details on maintaining the feedwater system. Key references include:

- Less than 4 hour Shut-down– see pages 4-3 to 4-7

Clayton Industries Steam Generator & Fluid Heater Installation Manual

This manual contains details on installing the system and components including Piping and Instrumentation (P&ID) diagrams.

Clayton Industries Steam Generator Instruction Manual

This manual contains operation details including periodic and component maintenance information. Key references include:

- Flue Gas Recirculation (FGR) – pages 2-8 to2-9
- Programmable Logic Controller – pages 2-17 to2-18
- Alarm Messages – pages 2-21 to2-22
- Filling Procedure – pages 4-1 to4-2
- Initial Starting – pages 4-2 to4-4
- Shutdown Procedures – pages 4-4 to4-8
- Troubleshooting – pages 5-1 to5-9
- Feedwater Pump Relief Valve – pages 7-9 to7-10
- Feedwater Pump Discharge Snubber – pages 7-33 to 7-36
- Feedwater Pump Belt Replacement – pages 7-33 to 7-34
- Blower Motor Belt Replacement – pages 7-34to 7-36

C.B. Boiler Overview and Procedures

Introduction

The Cleaver-Brooks FLX model boiler (shortened to C.B. Boiler for the rest of this manual) is a five-pass water-tube package boiler. The unit operating at the CCM is fueled solely by natural gas. The boiler consists of the formed tubes, the external down comer, and the top and bottom drums to which they connect. The heated area of the pressure vessel is contained within a gas tight insulated casing that is composed of removable formed steel panels.

It is required that CCM plant operators be familiar with the C.B. Boiler's Operating and Maintenance Guide, which covers the basics of operation and comes together with this copy of the CCM Operators' Manual.

This section of the CCM manual will go over the unique modifications made to this boiler for CCM operations and will discuss the Cleaver-Brooks HAWK 1000 controller.

C.B. Boiler HAWK 1000 Control System

The C.B. Boiler at the CCM is outfitted with a HAWK 1000 controller featuring a graphical Human-Machine-Interface (HMI) that displays boiler parameters, fault annunciation and alarm history, as well as providing access to boiler configuration and control functions.

The HAWK 1000 Boiler Control System provides boiler firing rate control to maintain steam pressure on set point. Final control element(s) are modulated via Modbus communications network to insure that the optimum fuel/ air ratio is maintained throughout the firing range.

The Hawk 1000 can be monitored by Building/Plant Automation Systems via an optional Cleaver-Brooks PT protocol translator. Ether Net/Internet communication also enables remote monitoring of the Hawk 1000 Boiler Control System.

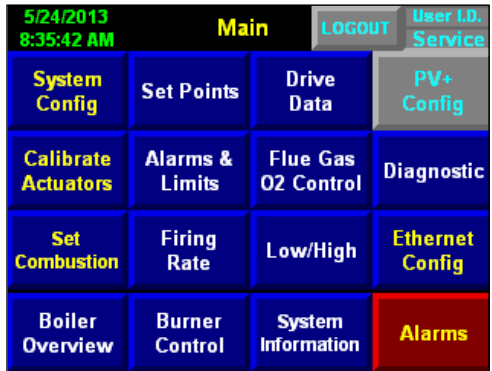
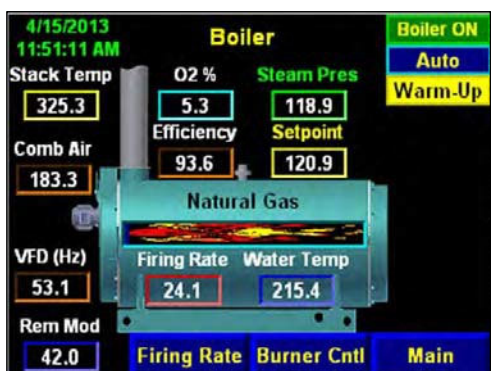
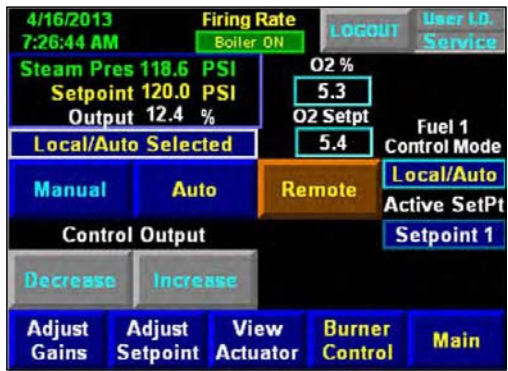
It is required that CCM operators be familiar with the Cleaver-Brooks HAWK 1000 Boiler Controls Manual, which comes together with this copy of the CCM Operators' Manual.

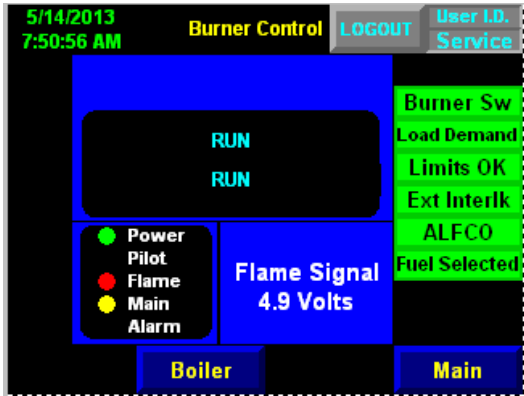
This section detailing the CCM C.B. Boiler will make reference to common features and operations of the HAWK 1000 controller.

Key Menus

The Cleaver-Brooks Hawk 1000 control interface has multiple menus and sub-menus. This section of the manual will describe the menus that are commonly used in the operation of the C.B. Boiler.

For information and advanced operation regarding the other menus, including the commissioning menus, refer to the Cleaver-Brooks Hawk 1000 Boiler Controls Manual.

Menu and Menu Description	Screenshot
Main Menu The Main Menu of the Hawk 1000 control interface is the highest in the menu hierarchy. From here, the operator can access all the main operating menus.	
Boiler Overview The Boiler Overview menu displays the real-time operating variables of the boiler, including firing rate, steam drum pressure, set point, and whether the boiler is in auto or manual mode. There are shortcut buttons to the Firing Rate, Burner Control menus, and Main menus.	
Firing Rate The Firing Rate menu allows the manual control of the boiler's burner output. This is also where the operator will switch the boiler firing rate controller into automatic.	

Menu and Menu Description	Screenshot
Burner Control The Burner Control menu is a graphic representation of the boiler's burner management system. During startup, the operator can view this menu to follow the boiler's startup and quickly troubleshoot any permissives that are not satisfied.	
Alarm Overview The Alarm Overview menu allows the operator to view, silence, and acknowledge any alarms currently active on the C.B. Boiler. If an alarm occurs during operation of the Cleaver-Brooks, the Hawk 1000 will automatically display this menu. This menu also provides a history of alarms.	

Cleaver-Brooks Drum Level and Boiler Feedwater Control Strategy

The C.B. Boiler, as of August 2016, operates using an on-off drum-level control strategy.

The drum-level is sensed by the Cleaver-Brooks Level Master Water Level Controller (herein called the Level Master). The Level Master's sensing element is an aluminum float and is housed in the water column on the right of the steam drum. This water column is also equipped with its own gauge glass.

When the drum level, measured in inches, drops to a pre-programmed value, the Level Master will start the feedwater pump. The feedwater pump draws water off the booster pump discharge header and pumps it into a dedicated feedwater line connected to the down-comer on the back of the C.B. Boiler. There is also an electrically-operated solenoid valve in this line that opens when the feedwater pump is running and closes when the pump shuts down.

Unlike the Clayton Boiler, there is no recirculation of water back to the Hotwell tank. Draining hot water from the boiler requires the operation of the blow-down valves, which discharge the boiler water into the blow-down tank. The blow-down tank tempers the water before it is dumped into the sewer. There is a drain valve on the blow-down line near the boiler that can send the boiler water straight to the sewer but this should only be done when the boiler is cold and being drained for its yearly inspection.

Level Master Details

The primary duty of the Level Master is to act as a low-water-cutout (LWCO) to protect the boiler from a dangerously-low water condition. The Level Master will trip the boiler if the water level becomes dangerously low. The Level Master can also sense a dangerously-high drum level and will trip the boiler if the drum level becomes too high.

The secondary purpose of the Level Master is to maintain the boiler feed level. As mentioned, this is done through on-off drum level control.

The Level Master's controller has non-volatile memory and logs alarms and events for later review.

Sensing Element



The Level Master does not contain any moving switches or relays. It senses the drum level in the boiler with an aluminum float. The position of the aluminum float generates an electrical-current signal that varies depending on the position of the float.

LWCO Redundancy

The Level Master features a variety of redundancies, including a stuck-float alarm and boiler trip. For more information, refer to the Level Master manual.

Level Master Menu Operation

The front face of the Level Master microprocessor unit contains three buttons:

- The Mode button is used for setting preset values and commissioning the Level Master
- The Blow Down button for accessing operating functions AND for bypassing the LWCO outside of the Semi-Automated Blow Down Routine.
- The Menu-Reset button for diagnostic and history retrieval as well as executing operating routines such as blowing down the Level Master water column without tripping the boiler.

Level Master Semi-Automated Blow Down Routine

The Level Master is set to perform a blow down of its water column at prescribed intervals controlled by its microprocessor. However, it cannot do these blow downs automatically as it can endanger the boiler. Therefore, an operator must carry out the blow down following the prompts from the Level Master's microprocessor display.

When it is time for the blow down, the screen on the Level Master's microprocessor will prompt the operator to perform the blow down sequence.

For a step-by-step guide on performing a successful blow down on the Level Master, consult the Semi-Automatic Blow Down of the Level Master Water Column procedure of this manual.

Advanced Operation of the Level Master Controller

For advanced operations of the Level Master Controller, refer to The Cleaver-Brooks Level Master CBLME Water Level Control Manual.

C.B. Boiler and Auxiliaries Documentation

This section provides references to the C.B. Boiler, and auxiliary equipment, documentation pertaining to the CCM steam systems.

- The Cleaver-Brooks Model FLX Packaged Boiler Operation, Service and Parts Manual
- The Cleaver-Brooks Hawk 1000 Boiler Controls Manual
- The Cleaver-Brooks Level Master CBLME Water Level Control Manual
- The Grundfos CR, CRI, CRN, CRT Installation and Operating Manual

The Cleaver-Brooks Model FLX Packaged Boiler Operation, Service and Parts Manual

This manual outlines the following information in regards to the Cleaver-Brooks Model FLX Boiler:

- Basics of Flexible Watertube Operation
- ProFire™ V Burner
- Pressure Vessel Care
- Sequence Of Operation
- Adjustment Procedures
- Trouble Shooting
- Inspection And Maintenance
- The Cleaver-Brooks Hawk 1000 Boiler Controls Manual

The Cleaver-Brooks Hawk 1000 Boiler Controls Manual

This manual outlines the following information in regards to the Cleaver-Brooks Hawk 1000 Controller:

- General Overview of the System
- System Components
- Hardware Checkout
- System Configuration
- Commissioning
- Diagnostics and Troubleshooting
- System Parts

The Cleaver-Brooks Level Master CBLME Water Level Control Manual

This manual outlines the following information in regards to the Cleaver-Brooks Level Master Controller:

- Features and Specifications
- Installation and Initial Testing
- Sensitivity Tables (for fine tuning level control)
- Commissioning and Operation
- Software Configuration
- Troubleshooting and Parts
- Message Display Details

The Grundfos CR, CRI, CRN, CRT Installation and Operating Manual

This manual outlines the following information in regards to the Cleaver-Brooks' Grundfos feedwater pump:

- Preventative Maintenance
- Motor Maintenance
- Motor Lubrication
- Pump Lubrication
- Diagnostic Procedures for Troubleshooting

Auxiliary Systems

Emergency Power

A 660 kW emergency diesel driven generator (EDG) is provided to supply power to the Clayton Boiler and its auxiliaries during a power outage. The CCM EDG is housed in the main building just east of the boiler room; access is from the east entry roadway.

The electrical switchgear for the EDG and utility power is located in Electrical Room 1124; access is from inside the facility near the EDG.

The fuel supply for the diesel engine is contained in the base of the machine. A 660 US (550 Imperial) gallon full fuel tank will last 11.8 hours at full load.

Only qualified and trained staff are permitted to test the EDG.



Figure 9 - Emergency Diesel Generator

Air Compressor and Medical Suction Air

There is an air compressor and medical suction air unit located on the second floor (Mechanical Mezzanine). The air compressors are registered with BCSA and will require inspection by a BCSA Boiler Safety Officer every five years.

Appendix A –Definitions and Equipment List

Definitions and Abbreviations

Term	Description
ADV	Automatic Dump Valve
APC	Auxiliary Pressure Control (header pressure control mode; as opposed to boiler pressure control). Additional detail on page APC-3 of the Clayton Instruction Manual Appendix 'A'
Auto Fill	Feature on boiler causing it to automatically fill with water
Auto shut down	Auto shut down of the boiler which includes post purging ready for auto re-start and intermittent flow of water through the boiler.
AUX	Auxiliary Operating Mode (see also APC)
BMS	Building Management System OR Burner Management System
Coil type boiler	Coil type steam generators are also known as “once through boilers”, and do not have a discernable water level.
Cooling fan	Boiler cooling fan located at back of boiler and provides cooling air between the casing and the furnace.
DDC	Direct Digital Control – the generic name for the CCM steam plant control system.
Delayed Rate	Causes the Auto Dump Valve to open while the boiler is coming up to operating pressure and temperature. Additional detail on page APC-3 of the Clayton Instruction Manual Appendix 'A'.
FGR	Flue Gas Recirculation
Fill mode	Boiler is filling with water
FSSS	Flame Safety Supervisory System
Hotwell	Also called the feedwater storage tank
Manual Fill	Manual control filling rate by pressing up and down buttons on the OIU
OIU	“Operator Interface Unit” – the LCD screen and keypad
RPRE	Regulated Pressure Retaining Equipment. List of pressure regulation items regulated by the safety authority.
RUN mode	Boiler is set to respond to changes in pressure as opposed to being on stand-by.
Separator	Separates water and steam after the mixture leaves the coil section of the boiler
SOP	Standard Operating Procedure
STD / Standard	Primary Operating Mode – pressure control is set to control boiler pressure (as opposed to controlling header pressure)

Term	Description
Super-economizer	Optional equipment that provides added heating surface and increased heat transfer between the flue gases and the heating coil. Preheats water before entering main coil section.
TDS	Total Dissolved Solids

UBC CCM Regulated Pressure Retaining Equipment (RPRE) List		
Count	Tag	Name
1	HE-9	STM/HWH Exchanger
2	HE-10	STM/HWH Exchanger
3	HEX-8	STM/DWH Exchanger
4	N/A	Medical Air Receiver
5	ET-1	HWH Expansion Tank
6	ET-2	CHW Expansion Tank
7	ET-3	DHW Expansion Tank
8	ET-4	DHW Expansion Tank
9	AS-1	HWH Air Separator
10	AS-2	CHW Air Separator
11	WSHP-1	Water Source Heat Pump
12	WSHP-2	Water Source Heat Pump
13	WSHP-3	Water Source Heat Pump
14	WSHP-4	Water Source Heat Pump
15	WSHP-5	Water Source Heat Pump
16	DHWT-1	DHW Storage Tank
17	DHWT-2	DHW Storage Tank
18	CHWBT-1	CHW Storage Tank
19	HWBT-1	HWH Storage Tank
20	ET-5	DHW Expansion Tank
21	HEX-1	Heat Recovery Exchanger
22	HEX-3	Heat Recovery Exchanger
23	HEX-4	Heat Recovery Exchanger

UBC CCM Regulated Pressure Retaining Equipment (RPRE) List		
24	HEX-5	Heat Recovery Exchanger
25	HEX-6	Heat Recovery Exchanger
26	HEX-7	STM/DWH Exchanger
27	Note: HEX-2 – Not Used	
28	B-1	STM Boiler by Clayton Industries
29	B-2	Cleaver-Brooks Model FLX Water Tube Boiler
30		Large Bulk Sterilizer manufactured by Getinge
31		Rack Washer manufactured by Northwestern Better Built.
32		Tunnel Washer manufactured by Northwestern Better Built.
33		Tissue Digester manufactured by Biosafe Engineering

Appendix B – Clayton Boiler Controller Interaction

Steam Pressure control

The Clayton boiler features the following pressure controls:

Pressure modulation:

- The boiler controls steam pressure at set- point with approximately five-to-one turn-down. Reduced demand for steam causes the pressure to increase, and the firing rate is decreased. Increased demand for steam causes the steam pressure to decrease, and the firing rate is increased to satisfy demand. Additional details of the pressure modulation system are provided in section II of the accompanying Clayton Instruction Manual.

High pressure cut-out:

- If steam pressure exceeds the high pressure set point, there is a high pressure cut-out that stops the burner. Once the pressure decreases to the pre-set limit, the boiler goes through its purge and start sequence to return to pressure modulation. The high pressure cut-out is an operational limit, no reset is required.

First high pressure trip:

- If the high pressure cut-out fails to activate, a high pressure trip switch will activate and the burner flame is extinguished. Should this high pressure trip activate, local manual reset is required before boiler operation can be resumed. An alarm on the boiler and at the remote monitoring location will sound.

Second high pressure trip:

- If the first high pressure cut out fails to activate, a second high pressure trip switch will activate and will cause the burner flame to be extinguished. Should this high pressure trip activate, local manual reset is required before re-start is possible. An alarm on the boiler and at the remote monitoring location will sound. (This is being installed as part of best practices for the ESA.)

Boiler safety relief valves:

- The boiler is fitted with two safety relief valves to vent steam should the high pressure cutout and first and second high pressure trips fail. (This is part of the original equipment supply.)

Remote Shut-down:

- In addition to the above, remote shut down of the Clayton Boiler is possible from the UBC Power House Control Center, should the need arise. (This is provided as part of best practices for the ESA.)

Coil-type Built-in Release:

- The coil type boilers, with the safety feature of inherently low water volume, were specifically chosen for the Center for Comparative Medicine application. If the pressure continued to rise beyond the safety relief valve setting, the coil would eventually fail and steam would be released into the boiler furnace as demonstrated on the Clayton video at this link: http://claytonindustries.com/clayton_v1_videos.html.

Clayton Boiler Steam Separator High Level Trip

- The steam separators are fitted with a high level trip switch. In the event of a high water level in the steam separator, the boiler will trip.

Note: The boiler has a minimal water volume and cannot produce a catastrophic explosion as is typical of firetube boilers.

Clayton Boiler Temperature control

There are four levels of steam temperature control. Forced circulation of water through the boiler is programmed via the VFD's to follow the combustion air and natural gas flow curves. The boiler is designed to maintain temperatures within design limits by maintaining a flow of feedwater through the coil. The feedwater pump for the Clayton Boiler is individually controlled by dedicated VFDs, which are controlled by the Clayton Boiler's OIU.

The four levels of steam temperature control are as follows:

First level - Auxiliary Temperature Sensor (ATS) / trip switch:

- Located on the steam separator. If the temperature in the separator were to rise above that preset limit, the boiler would trip on high separator temperature and the gas burner flame would be extinguished. This sounds an alarm and local manual reset is required prior to restart. Details of this feature are found in section 2.2.4.4 of the accompanying Clayton Instruction Manual.

Second level – A second Auxiliary Temperature Sensor (ATS) / trip switch is also provided:

- This sensor for the switch is also located on the steam separator while the switch is mounted inside the boiler control panel. If the temperature in the separator were to rise above that preset limit, the boiler would trip on high separator temperature and the gas burner flame would be extinguished. This sounds an alarm and local manual re-set is required prior to restart.

Third level - high temperature switch MTLC1:

- Located in the coil as detailed in 2.2.4.1&2 of the accompanying Clayton Instruction Manual. If the coil temperature were to rise above the preset limit, the boiler would trip on high coil temperature and the gas burner flame would be extinguished. This sounds an alarm and local manual reset is required prior to re-start.

Fourth level - high temperature switch MTLC2:

- Located in the coil as detailed in 2.2.4.1&2 of the accompanying Clayton Instruction Manual. MTLC2 is set at a higher temperature than MTLC1. If the coil temperature were to rise above the preset limit, the boiler would trip on high coil temperature and the gas burner flame would be extinguished. This sounds an alarm and local manual reset is required prior to re-start.

Note: The first indication of high temperature is usually detected by MTLC1 or MTLC2 as those sensors are in the coil inside the boiler while the others are in the steam separator.

Clayton Boiler Feedwater Control

There are five levels of feedwater control:

First level - variable frequency drive (VFD):

- Feedwater flow is pre-programmed to vary with combustion air flow and natural gas flow as the steam demand changes.

Second level - Auto Dump Valve (ADV) system:

- On start-up, pre-programmed settings open a solenoid operated valve that allows water from the separator to return to the hotwell and the boiler is kept at minimum firing rate until the proper temperature, pressure, and time settings are achieved.

Third level- steam trap on the separator:

- Water entering the separator flows to the steam trap and is discharged to the hotwell for re-circulation.

Fourth level - high level switch in the separator:

- Should the water level rise in the separator to the level of the switch, the high-level trip would activate and the gas burner flame would be extinguished. This sounds an alarm and manual re-set is required prior to restart.

Fifth level:

- Should the hotwell water level go low, two separate low level probes have been fitted on the hotwell, will trip and shut down the boiler.

Note: A low level in the hotwell is indicated by a red light on the hotwell level switch panel. It will be illuminated prior to the switch causing the boilers to trip.

Clayton Boiler Flame Safety Supervisory System

The Clayton Boiler is fitted with a flame safety supervisory system (FSSS). The FSSS includes high and low gas pressure interlocks, flame scanner, and combustion air pressure sensor. The FSSS is detailed in section VI of the accompanying Clayton instruction manual.

Parameters for automated safety shut-down

Pressure settings

The **modulating pressure set point** is at **90 PSIG** (As of August, 2016)

The **high pressure cut out** is set at **130 PSIG** (As of August, 2016)

The **first high pressure trip** is set at **135 PSIG** (As of August, 2016)

The **second high pressure trip** is set at **140 PSIG** (As of August, 2016)

The steam safety relief valves on the Clayton Boiler are set at 150 PSIG (As of August, 2016)

Temperature settings

Separator high temperature switch #1 is 380°F and #2 is 400°F.

MTLC1 for high coil temperature is 475 - 575 °F.

MTLC2forhigh coil temperature is 500 - 600°F

Clayton Boiler Fail Safe Conditions

The following measures result in a fail-safe condition should there be a failure in the control system:

1. **Modulating pressure control system failure:** the high pressure cut out will take effect on rising pressure.
2. **High pressure cut-out failure:** the first high pressure trip will take effect on rising pressure.
3. **First high pressure trip failure:** the second high pressure trip will take effect on rising pressure.
4. **Second high pressure trip failure:** the safety valves will open on rising pressure.
5. **Temperature trip switch failure:** each are wired individually; if any fail, the others remain in effect.
6. **Open circuits:** a circuit open in any control system will cause the safety feature (i.e.: a high pressure safety switch) to come into effect.
7. **Power loss to the controls:** the gas burner flame will be extinguished. This will sound an alarm and manual re-set will be required prior to restart.

Appendix C – C.B. Boiler Controller Interaction

Steam Pressure control

The C.B. Boiler features the following pressure controls:

Pressure modulation:

The C.B. Boiler controls steam pressure at set-point programmed into the Hawk 1000 Control System. Reduced demand for steam causes the pressure to increase and the firing rate is decreased. Increased demand for steam causes the steam pressure to decrease and the firing rate is increased to satisfy demand. Additional details of the pressure modulation system are provided in The Cleaver-Brooks Hawk 1000 Boiler Controls Manual.

High pressure cut-out:

If steam pressure exceeds the high pressure set point, there is a high pressure cut-out that stops the burner. Once the pressure decreases to the pre-set limit, the boiler goes through its purge and start sequence to return to pressure modulation. The high pressure cut-out is an operational limit, no reset is required.

High-High pressure trip:

If the high pressure cut-out fails to activate, a high pressure trip switch will activate and shutdown the burner. Should this high pressure trip activate, local manual reset is required before boiler operation can be resumed. An alarm on the boiler and at the remote monitoring location will sound.

Boiler safety relief valves:

The boiler is fitted with one safety relief valve to vent steam should the high pressure cutout and first and second high pressure trips fail.

Remote Shut-down:

In addition to the above, remote shut down of the boiler is possible from the UBC Power House Control Center, should the need arise. (This is provided as part of best practices for the ESA.)

C.B. Boiler Steam Drum Water Level Trips

The steam drum is equipped with two water columns, located on either side of the steam drum and a water level sensing probe; the controls are located on top of the steam drum with the probes in the water space of the steam drum.

Low-water-level cut-outs (LWCO) functions are provided by the water sensing probe and by the two water columns. One water column houses an auxiliary LWCO and the other water column houses the Cleaver-Brooks Water Level Master drum level controller.

The Level Master drum level controller acts as both a high- and low-water-level cut-out.

The auxiliary LWCO acts as a backup to the Level Master and the water level sensing probe. Both the auxiliary LWCO and the water sensing probe must be reset if tripped.

C.B. Boiler Feedwater control

The C.B. Boiler operates on an on-and-off feedwater control strategy.

The Cleaver-Brooks Level Master controller monitors the level in the boiler steam drum. When the drum level drops to a preset cut-in value, the Level Master will start the Grundfos feedwater pump and open the solenoid valve in the feedwater line to the boiler.

Once the drum level is raised to a preset value, the Level Master will shutdown the feedwater pump and closes the solenoid valve.

The on-and-off operation of the feedwater pump occurs intermittently during boiler operation. The pump will not start if the boiler drum level is above the cut-in value.

Boiler water “Shrink and Swell”

Level control in water tube boilers may be affected by ‘shrink and swell’ conditions.

Level shrink occurs when there is a sudden and major reduction in steam demand. The increase in steam drum pressure causes a reduction of water volume in the steam drum, possibly resulting in a false low-drum-level. The level controls may over-fill the boiler with feedwater.

Level swell occurs when there is a sudden major increase in steam demand. The decrease in steam drum pressure causes an increase in water volume in the steam drum, possibly resulting in a false high-drum-level. The level controls may starve the boiler of feedwater.

Effective management of ‘shrink and swell’ may require additional training and expertise.

Hotwell Level Trip:

Should the hotwell water level go low, a low level switch, fitted on the hotwell, will trip and shut down the boiler.

Note: A low level in the hotwell is indicated by a red light on the hotwell level switch panel. It will be illuminated prior to the switch causing the boilers to trip.

C.B. Boiler Flame Safety Supervisory System

The C.B. Boiler is fitted with a flame safety supervisory system connected to the Hawk 1000 Controller.

Flame system interlocks and permissives include:

- The Low gas pressure switch
- The high gas pressure switch
- No-flame signal from the flame scanner (flame-loss)
- The burner on-off switch
- Main gas valve auxiliary switch
- Blower Motor Starter circuit
- Combustion air proving switch
- Low-fire proving circuit
- High-fire proving circuit

These interlocks and permissives are detailed in The Cleaver-Brooks Model FLX Packaged Boiler Operation, Service and Parts Manual.

C.B. Boiler Parameters for automated safety shut-down

Pressure settings

The **modulating pressure set point** is at **100PSIg** (As of August 2016)

The **high pressure cut out** is set at **120PSIg** (As of August 2016)

The **low pressure cut in** is set at **105PSIg** (As of August 2016)

The **first high pressure trip** is set at **135 PSIg** (As of August 2016)

The **second high pressure trip** is set at **140 PSIg** (As of August, 2016)

C.B. Boiler Fail Safe Conditions

The following measures result in a fail-safe condition should there be a failure in the control system:

1. **Modulating pressure control system failure:** the high pressure cut out will take effect on rising pressure.
2. **High pressure cut-out failure:** the first high pressure trip will take effect on rising pressure.
3. **First high pressure trip failure:** the second high pressure trip will take effect on rising pressure.
4. **Second high pressure trip failure:** the safety valves will open on rising pressure.
5. **Open circuits:** a circuit open in any control system will cause the safety feature (i.e.: a high pressure safety switch) to come into effect.
6. **Power loss to the controls:** the gas burner flame will be extinguished. This will sound an alarm and manual re-set will be required prior to restart.

Appendix D - CCM Boiler Room Safety Equipment

Carbon Monoxide Monitor

A carbon monoxide (CO) monitor is located in the steam boiler room. A high CO reading will initiate local and remote audible and visual alarms. There is a push-button outside the CCM steam boiler room that can be pressed to silence the alarm.

Fire Extinguishers

A variety of fire extinguishers, including carbon-dioxide and dry chemical, can be found throughout the CCM plant and facility.

Fire Alarm Pull Stations

Fire alarm pull stations can be found by major entrances in the CCM plant and facility.

CCM Steam Boiler Emergency Shutdown Buttons

There are two CCM steam boiler emergency shutdown push-buttons located outside the room where the steam boilers are located. Each boiler has its own shutdown switch.

The emergency shutdown buttons should be tested at least once per month.

CCM Steam Boiler Alarm Strobe Lights

The boiler alarms are tied into strobe lights located inside the CCM plant and outside the CCM facility. These lights will flash if the CCM steam boilers are in an alarm state.

CCM Fire Alarm Strobe Lights

The CCM facility fire alarms are tied into strobe lights located inside the CCM plant and outside the CCM facility. These lights will flash if the fire alarms system for the CCM facility has been activated.

Appendix E - CCM DDC Pages

The following is a collection of key pages from the DDC monitoring system.

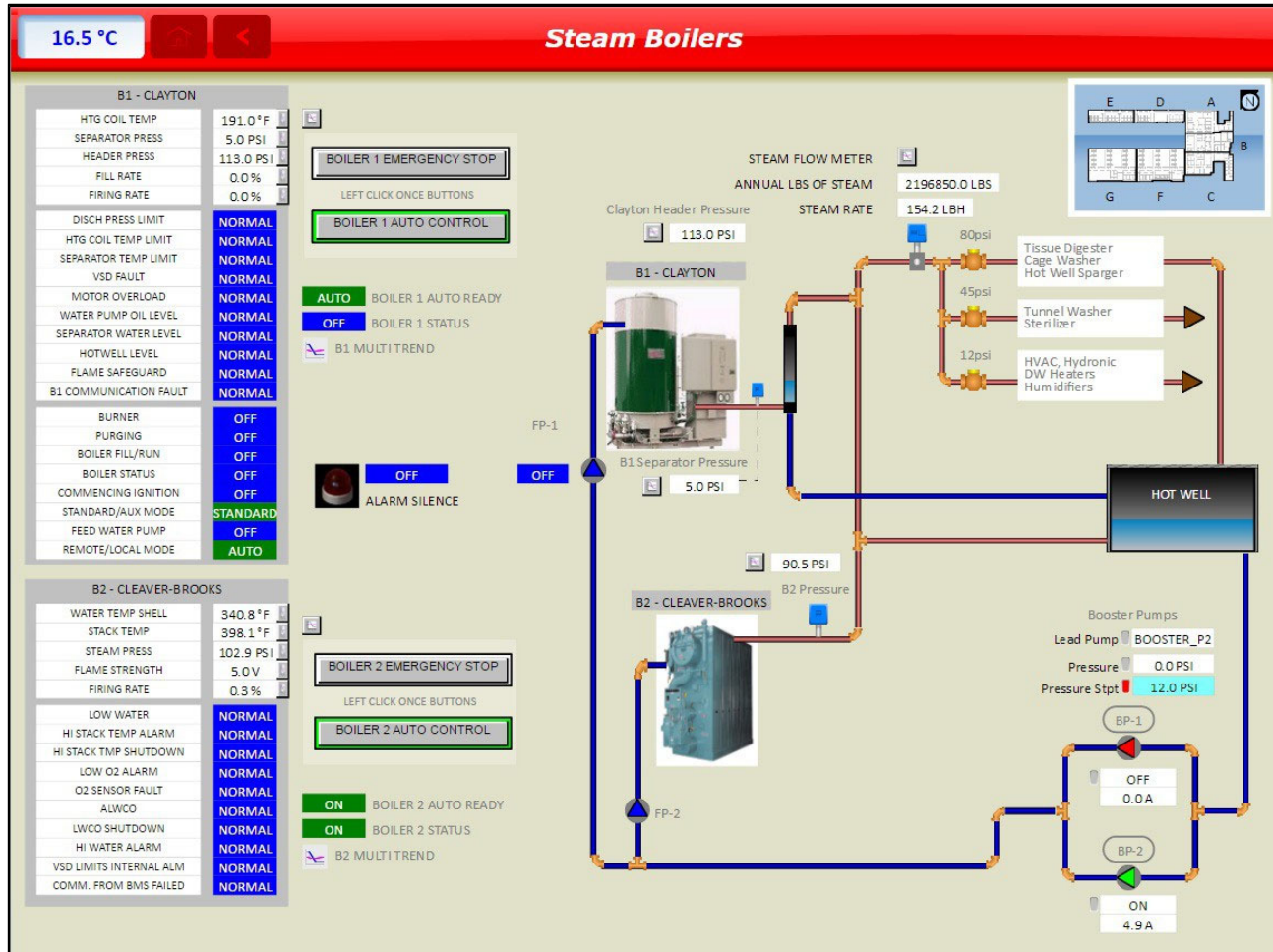


Figure 10 - Steam Boilers Page

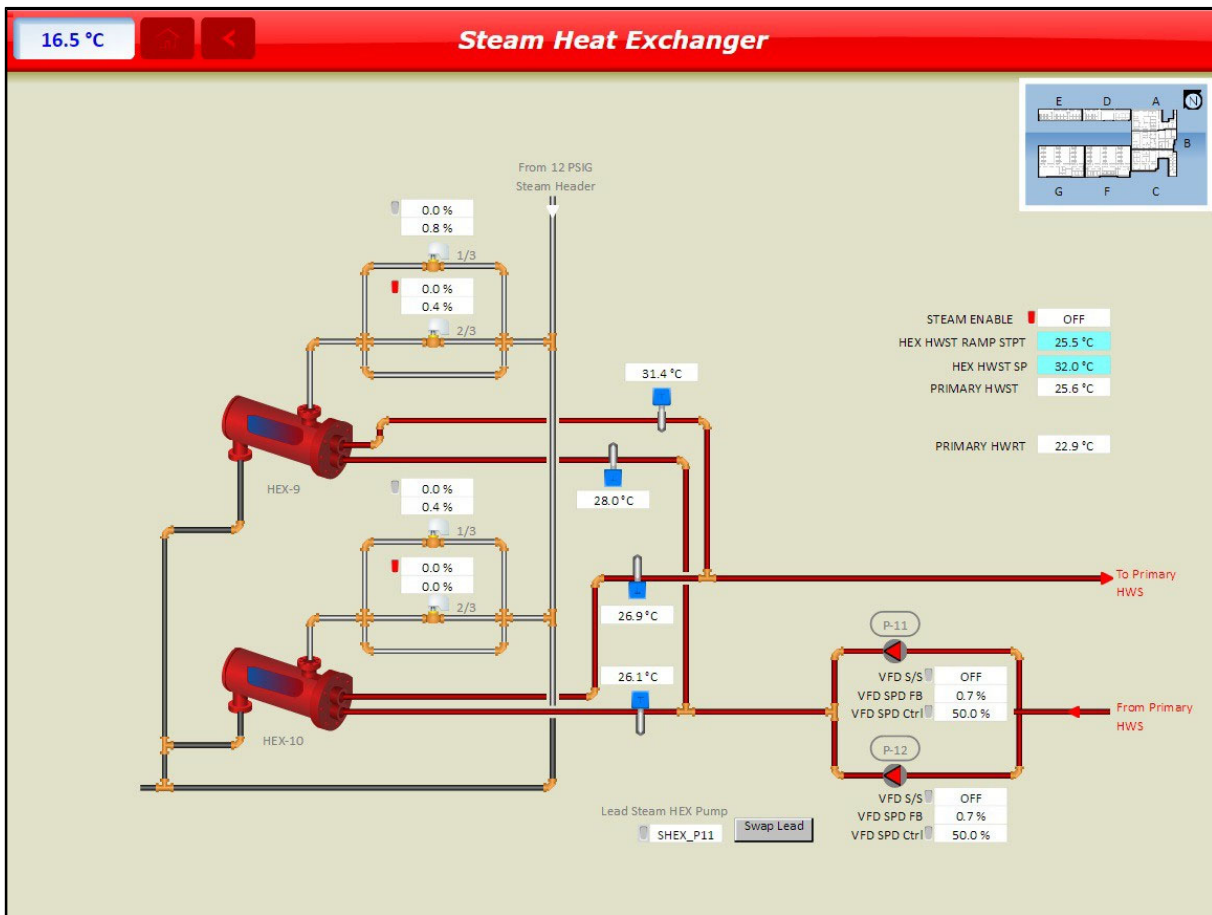


Figure 11 - Steam to domestic hot water heat exchangers page

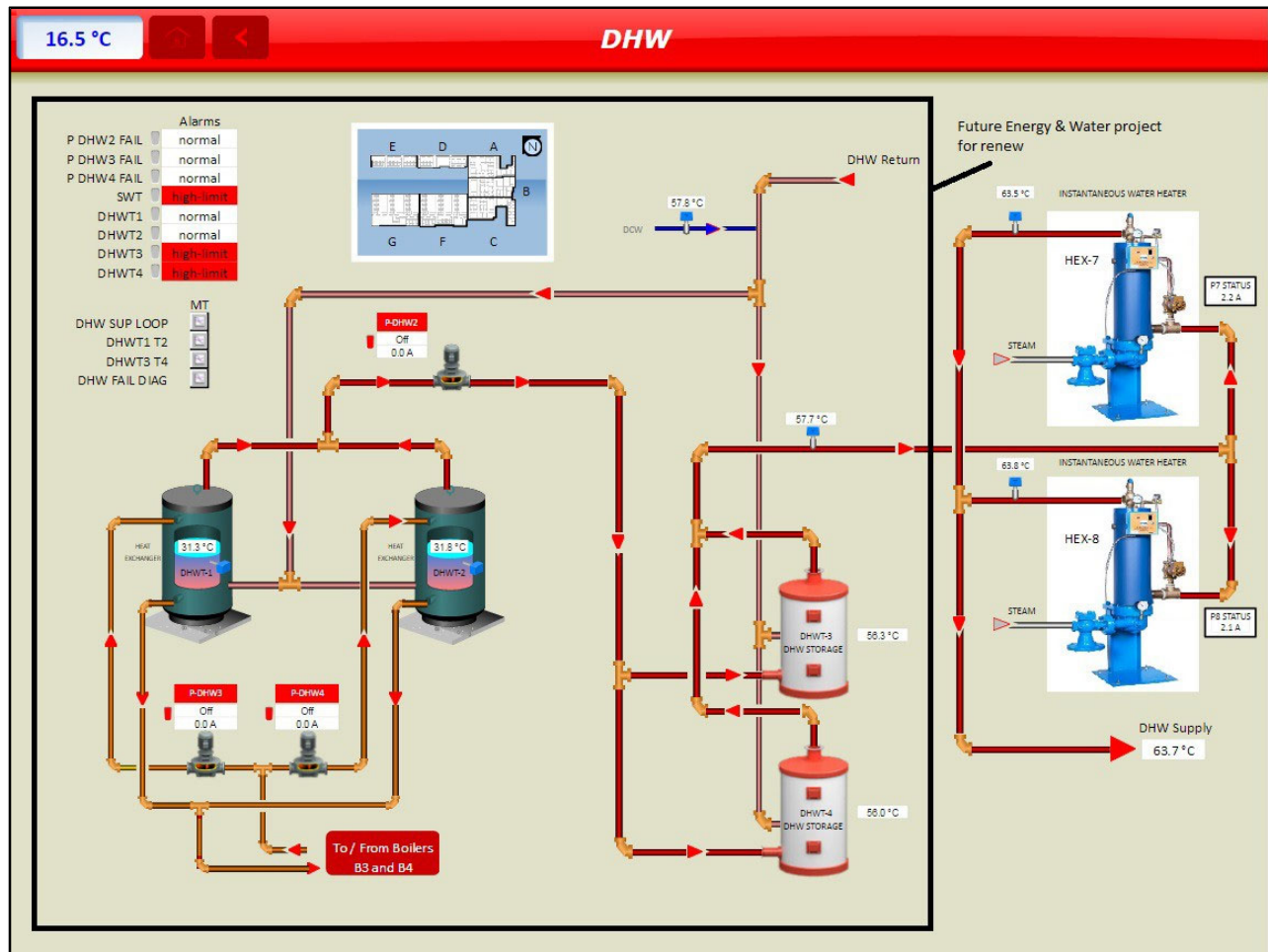


Figure 12 - Domestic Hot Water boiler page

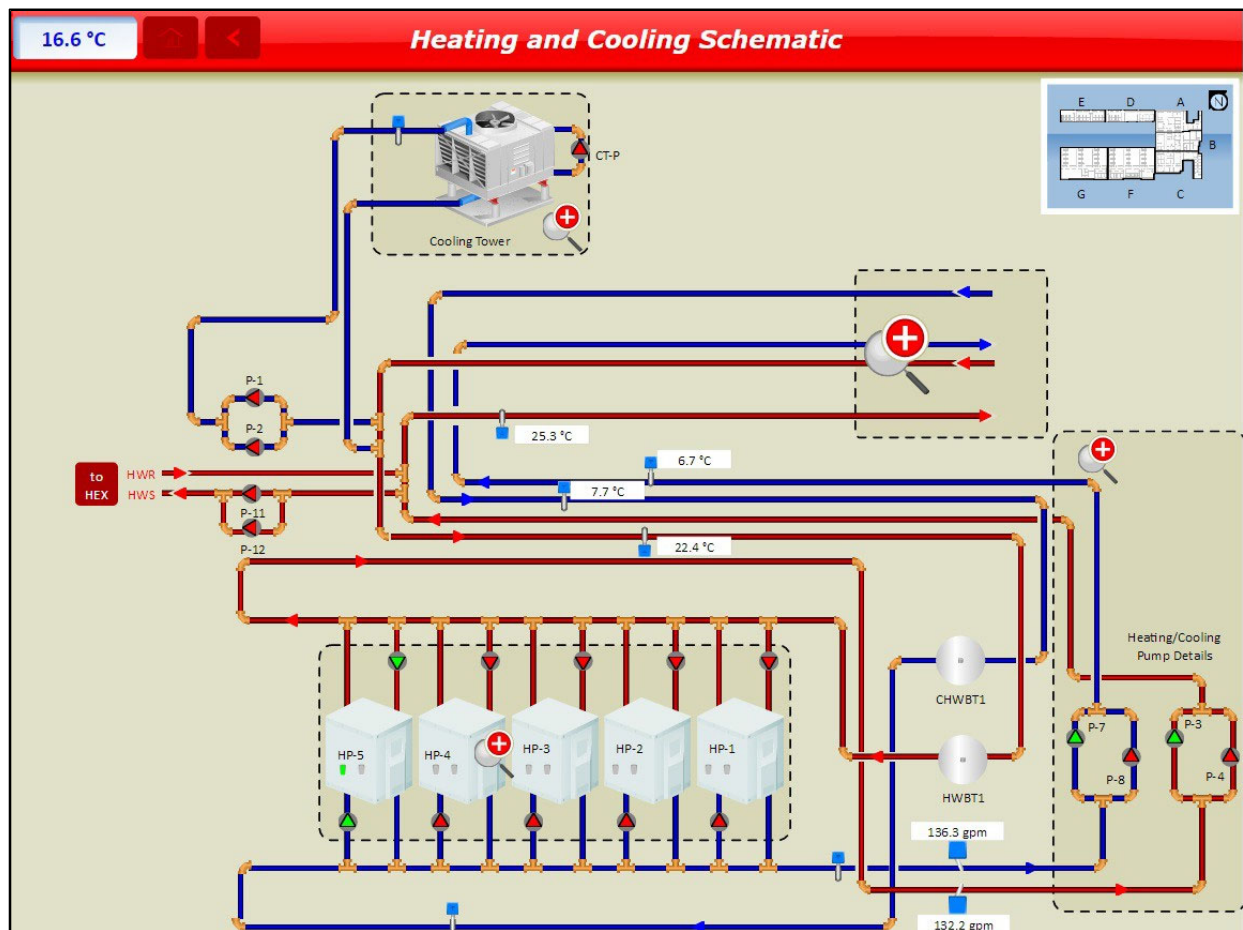


Figure 13 - Heating and Cooling Schematic page

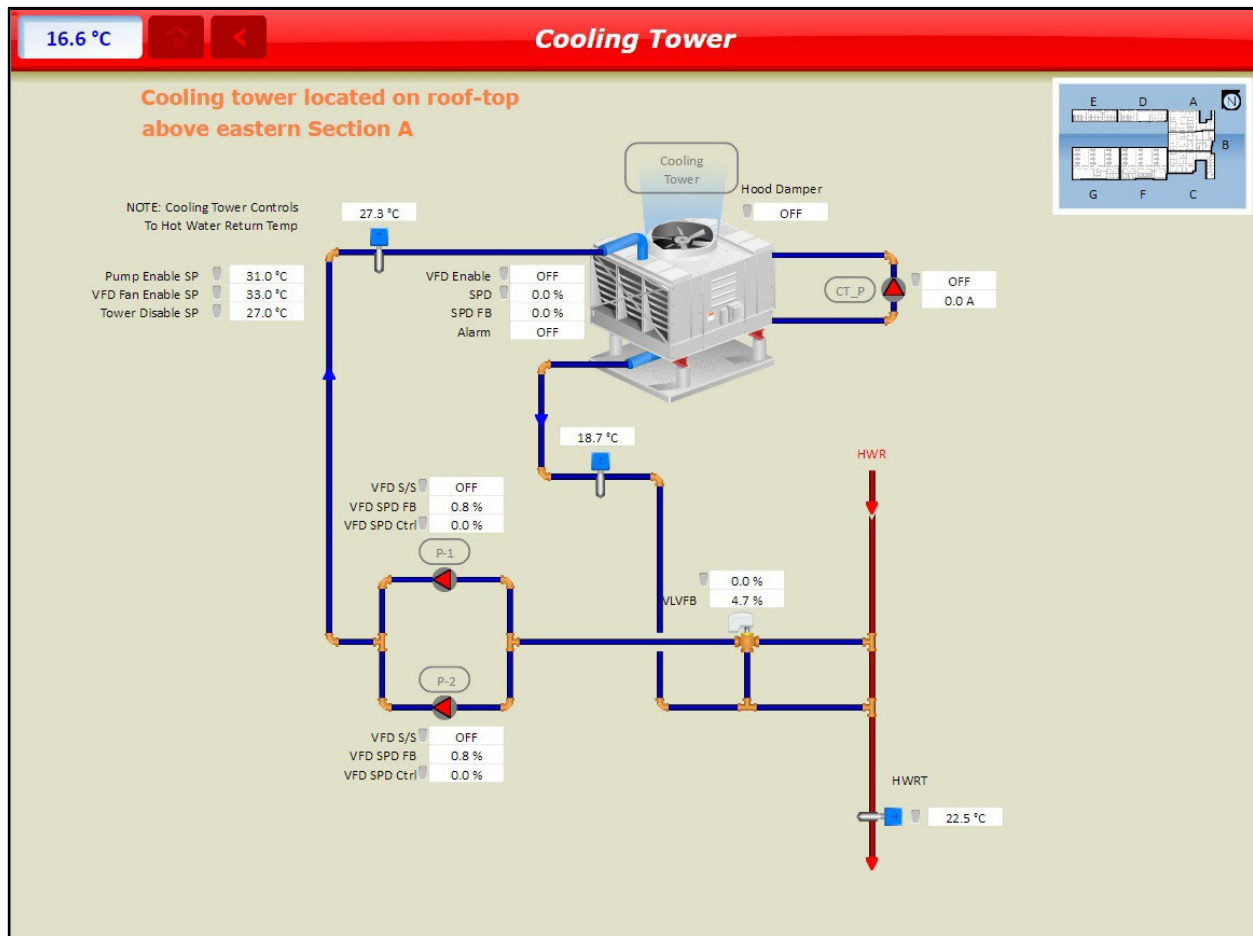


Figure 14 - Cooling Tower page

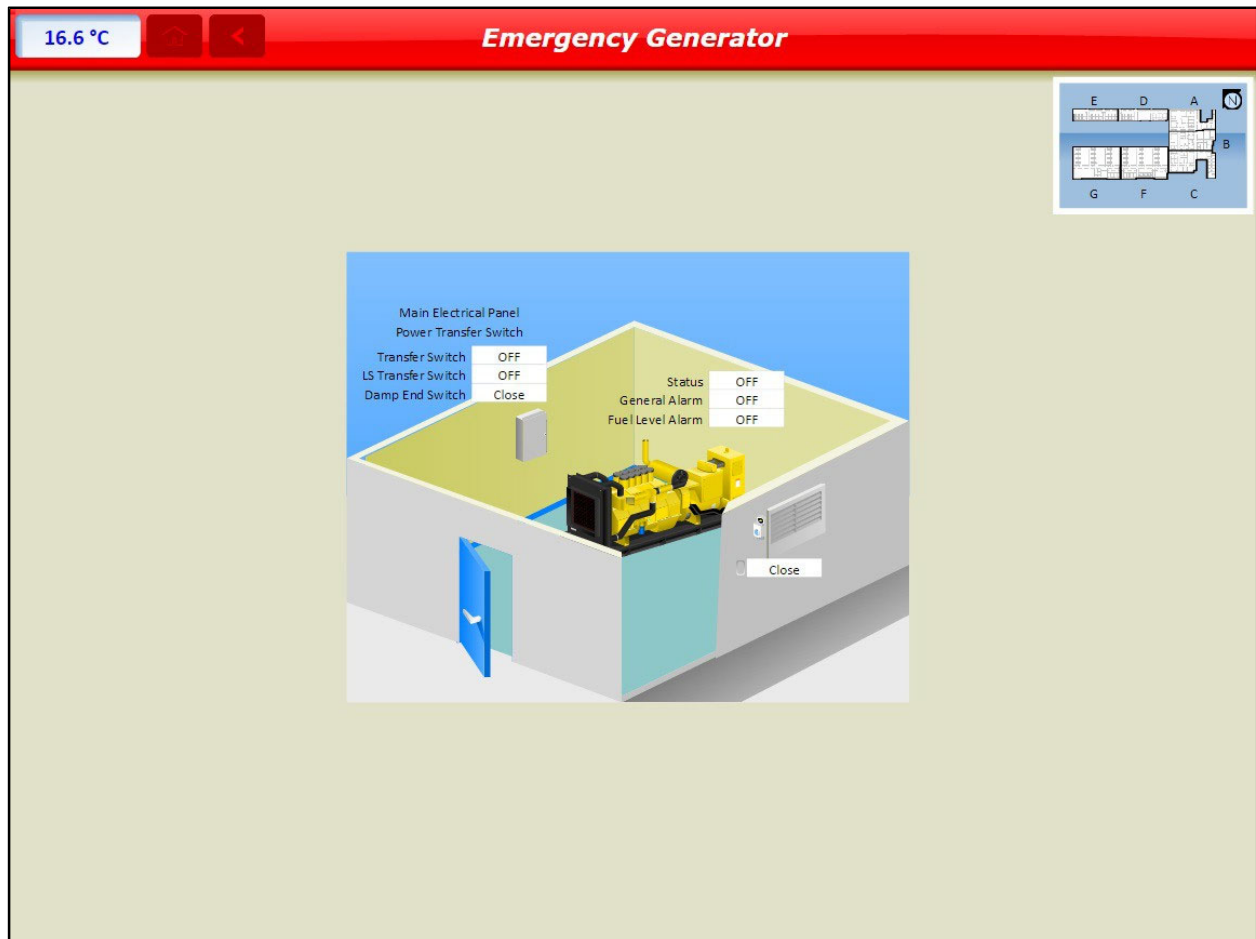


Figure 15 - Emergency Generator page

End of Manual

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